

Population growth of a gorgonian coral: equilibrium and non-equilibrium sensitivity to changes in life history variables

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Summary. A size dependent model of population growth of the Caribbean gorgonian *Plexaura A* is developed based on observed rates of survival, growth and colony fragmentation at a site in the San Blas Islands, Panama. Sensitivity and elasticity analyses indicate that the fate of large colonies has the greatest effect on population growth. Variables which directly affect the generation of large colonies have the next greatest effect on population growth. These variables include the recruitment of large fragments, and the survivorship of colonies in the next smaller size class. Sexual reproduction has an extremely limited ability to affect population growth. Vegetative reproduction has a greater potential effect on growth rates. Environmental conditions regularly change the matrix of transition probabilities which predicts population growth. This keeps the population from approaching its stable size class distribution. Deviations from the stable size class distribution alter sensitivity and elasticity and in this case have the effect of increasing the importance of survivorship of the smallest colonies. Non-equilibrium conditions alter sensitivity analyses and it is important to assess whether populations are at equilibrium and to determine the effects of such deviations on the sensitivity analysis.

Over the past 15 years the role and importance of clonal growth has received increasing attention in the study of plant and animal population ecology. This attention can be attributed to the recognition that clonal growth is common in many communities (Harper 1977; Jackson 1977) and that it significantly alters the manner in which selection operates on a population. Clonal growth affects features such as size and form (Coates and Jackson 1985; Lovett Doust 1981; Ryland and Warner 1986; Sebens 1981), longevity (Hughes and Jackson 1980), reproduction (Harper 1977; Highsmith 1982; Jackson and Coates 1986; Sebens 1981), competition (Jackson 1977), susceptibility to disturbance (Highsmith 1982; Jackson 1985)

and perhaps even rates of evolution (Jackson and Coates 1986).

Paralleling these efforts has been the development of increasingly sophisticated models which quantify population growth among clonal organisms (Hughes 1984; Hughes and Jackson 1985; Caswell 1985). These models have their origins in the modelling of complex life cycles (Lewis 1976, 1977) and size-based demographic models (Caswell 1982a, b; Hubbell and Werner 1979; Hughes 1984; Werner and Caswell 1977). The models are based on the distribution of individuals among different size classes and the probabilities of an individual moving from one size class to another in a given time. These probabilities form a matrix of transition probabilities, termed the population matrix. The product of the population matrix and the vector describing the population size structure yields the population size structure after one time unit. Iterative repetition of this process allows one to calculate both population size structure and population growth over time. This modelling procedure can be used for open populations (i.e. populations with recruitment which is independent of population size) by making annual adjustments for recruitment not otherwise included in the model (c.f. Hughes 1984). When the model has a completely closed structure, population growth and size structure can be determined analytically. Caswell (1982a, 1985) describes procedures for the analytic solution of these Markovian models.

Population growth models of these types are particularly useful in assessing the effects of altered environmental conditions or life history traits because they translate changes in probabilities of survival and growth into a common unit, population growth. Such analyses are qualitatively different from those which compare life history characters without reference to their quantitative effect on population growth. For instance, Weinberg et al. (1986) measured the effects of different experimental treatments on growth and survival of the bivalve *Gemma gemma*. Their treatments produced significant differences in the populations but their analysis revealed that the changes would have little effect on the growth trajectories

of the populations. This result may also be deduced by calculating the sensitivity and elasticity of the elements of the population matrix. Sensitivity is the absolute change in growth rate per unit change in the transition probability, and elasticity is the relative change in growth rate divided by the relative change in the transition probability. The magnitude of the sensitivities and elasticities identifies those size classes and transitions whose fates have particularly great effects on population growth (Caswell 1982 a, b; deKroon et al. 1986; Weinberg et al. 1986).

Sensitivity and elasticity analyses have been used to determine the effects of experimental treatments (Weinberg et al. 1986), to compare alternative life history strategies (Caswell and Werner 1978; Levin et al. 1987), to estimate the effects of different individual growth strategies on population growth (Ebert 1985), and to identify life history stages which most affect population growth (Caswell and Werner 1978; deKroon et al. 1986). Such analyses provide a quantitative basis for identifying the potential for selection to operate on a given aspect of an organism's life history. However, these analyses assume steady state environmental conditions and equilibrium population structure. Since many populations are subject to substantial environmental change it is also important to determine the extent to which sensitivity analyses are dependent on equilibrium populations. In this paper I present a size dependent model of population growth of a fragmenting gorgonian coral, and conduct a sensitivity and elasticity analysis to identify those steps in the life history that most affect population growth. Using non-equilibrium estimates of sensitivity and elasticity I compare analyses based on equilibrium solutions to those which occur when the population is far from its stable size class distribution.

Two types of non-equilibrium condition are considered. Firstly as environmental conditions change the population matrix, and thus the stable size structure of the population, changes. For instance, Hughes (1984) predicted different population size structures for Jamaican *Agaricia agaricites* using population matrices from storm and non-storm years. Changes in the population matrix prevent the population from reaching the stable size class distribution, which affects both population growth rate and the sensitivity analysis. The magnitude of this effect will be great when the environment is subject to frequent and large changes, and the effects will be small when the environmental fluctuations are small and/or gradual.

A second type of non-equilibrium condition is due to short-lived events such as pulses of recruitment or catastrophic mortality events. Many reef anthozoans have life histories with rare and sporadic recruitment and/or recruitment which is independent of the local population (Bak and Engel 1979; Rylaarsdam 1983; Wallace 1985). More dramatic changes in population size and structure occur as a result of extreme events such as hurricanes (Woodley et al. 1981), coral bleaching (Glynn 1988; Coffroth et al. 1989) and *Acanthaster* outbreaks (Pearson 1981). These events dramatically alter the size class distribution of the population, but the population then re-

covers under conditions in which the population matrix remains constant (but see Knowlton et al. 1990). Again the effects of such events are dependent on their frequency and magnitude.

Methods

The species modelled in this study is an undescribed plexaurid gorgonian, *Plexaura* A, which is common in the San Blas Islands of Panama and is known from throughout the Caribbean. Lasker (1984) describes many of the traits of *Plexaura* A. As in all clonal species *Plexaura* A population dynamics are confused by multiple levels of "individuality." One may consider the genetic unit, the clone or genet in the usage of Harper (1977), or the physiological and ecological unit, the colony, or the simplest unit, the polyps which make up the colony. In this paper I discuss the demography of colonies. Analysis at this level is in part dictated by the methodological difficulties of distinguishing and following either clones or polyps. Use of the colony as the basic unit of measure also reflects the status of colonies as the pertinent "ecologic unit." Numbers of colonies should more accurately characterize species abundances and the interactions among species than would numbers of either clones or polyps.

Plexaura A demographics can be divided into processes of recruitment, survivorship and growth into larger or shrinkage into smaller size classes. These processes and the *Plexaura* A population growth model are diagrammed in Fig. 1. In the model the *Plexaura* A population is divided into three size classes (0–10 cm, 10–20 cm and > 20 cm). Associated with each size class are three transition probabilities, which characterize the probability of a colony surviving and either remaining within the same size class or growing/shrinking into one of the other size classes. The transition probabilities make up a matrix, the population matrix, in which $P(2,1)$ denotes the probability of a colony in size class 1 growing into size class 2 in a 1-year period.

The probabilities of survival and growth used in the model are based on survivorship, growth and recruitment data observed on a 0.5 × 5-m belt transect at Korbiski Reef in the San Blas Islands, Panama, between 1984 and 1988. A total of 264 attached colonies were monitored and annual probabilities of surviving and moving from one size class into another are based on the histories of those colonies (Lasker 1990). The survival and growth of the colonies on the transect was controlled by mortality due to holdfast failure, size dependent growth rates, and reduction in size or mortality due to the grazing activities of the snail *Cyphoma gibbosum* and the polychaete *Hermodice carunculata* (Lasker 1990).

Reproduction in the *Plexaura* A population is divided into planula production and vegetative propagation. As in all gor-

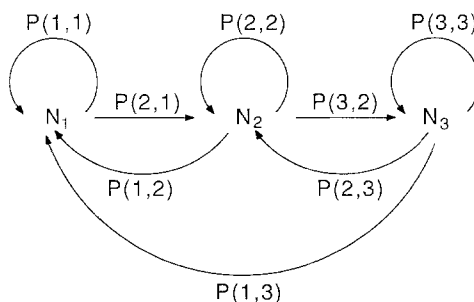


Fig. 1. Model of population growth for *Plexaura* A. Numbers of colonies in size classes 0–10, 10–20 and > 20 cm are denoted by nodes n_1 , n_2 and n_3 . The probabilities of moving into a different size class or remaining in the same size class are characterized by the product of the probabilities along any path between nodes. Movements between nodes require 1 year

gonians, *Plexaura* A colonies produce planulae (Brazeau and Lasker 1989). Although larval recruitment is rare (Lasker 1990) some larvae survive and presumably all recruitment to distant reefs occurs via larval recruitment. Unlike many gorgonians *Plexaura* A colonies can also develop vegetatively from fragments of established colonies (Lasker 1984). Large colonies produce fragments some of which reattach and form new colonies, and most population growth can be attributed to vegetative propagation (Lasker 1984, 1990). Thus *Plexaura* A reproduction is controlled by the rates of larval settlement, fragment production, and fragment survival.

In the model reproduction is restricted to colonies > 20 cm. The designation of 20 cm as the size at which reproduction starts is based on a combination of data on both fragment production and larval production. The distribution of scars from broken-off branches suggest that colonies smaller than 20 cm do not generate fragments (Lasker 1990). Similarly colonies less than 20 cm in height produce few if any mature gametes (Brazeau and Lasker 1989).

Vegetative reproduction contributes to elements $P(1,3)$, $P(2,3)$ and $P(3,3)$ of the population matrix. $P(3,3)$ is the sum of the probabilities of a colony surviving and not changing size plus the probability of producing a large fragment which successfully reattaches (recruits). $P(1,3)$ and $P(2,3)$ are the sum of the probabilities of a colony shrinking in size to the respective size class plus the probability of a large colony producing a vegetative recruit of the respective size.

The probabilities of producing vegetative recruits were derived from observed rates of fragment production and fragment survival. The number and size of unattached fragments found in a 100-m² study site (mean of 44 monthly collections (Lasker 1990) was used to calculate a per capita rate of fragment production. Total fragment production in each of the three size classes was adjusted to per capita rates by assuming that all fragments were produced by the 163 large colonies in the monitored 100 m² area. Although some transport of fragments was observed, the survey area was surrounded by other *Plexaura* A colonies and net export and import of fragments into the study quadrat are assumed to have been similar.

Fragment production rates were adjusted to recruitment rates using data on 1-year survival of tagged fragments in an adjacent 10 × 10-m plot (Lasker 1990). One-year survival of tagged 1-, 6-, or 18-branch fragments (approximately equivalent to (10, 10–20 and > 20 cm fragments) was used to estimate survival rates of fragments. Survival of small and intermediate sized fragments are averages of eight experiments (100 fragments per experiment) conducted between 1982 and 1988. Survival of the large fragments is based on four experiments conducted between 1985 and 1988. Fragment survival and reattachment was temporally variable in survivorship experiments, and 1-year survivorship of 6 branch fragments ranged from 0.0 to 0.23. Fragments with 6 branches tended to have higher 1-year survival rates than either 1- or 18-branch fragments (Lasker 1990). Since most fragments were in the 10–20 cm size class and those fragments had the greatest probability of survival, fragmentation had its greatest effect on $P(2,3)$.

Larval recruitment would contribute to $P(1,3)$ in the model. However, larval recruitment was not included, since it would require the assumption of a closed population. *Plexaura* A planulae float for at least several hours following their release (personal observation), suggesting that recruitment within reefs is not controlled by the local colony density (i.e. the population is open). Fortunately, the exclusion of larval recruitment has only minimal effect on the model due to the extremely low rates of larval recruitment. Only seven recruits were observed over 6 years, and larval recruitment rates were so low that assuming a closed population and including larval recruitment had no quantitative effect on the results.

Population growth

Equilibrium population growth rates were determined by calculating the principal eigenvalue of the population matrix. The stable

population structure was predicted by the associated eigenvector. Growth rates were calculated using data from each of the four years of data as well as that for an "average" year. Since larval recruitment was not included the model characterizes growth of an average clone. Growth rates were also predicted by taking the observed size class distribution and multiplying it by the observed population matrix. That calculation predicts subsequent size class structure if all of the assumptions of the model are correct and if the estimates of vegetative propagation are correct.

Sensitivity

Sensitivity is defined as $\delta\lambda/\delta P(i, j)$, where λ is the growth rate of the population and $P(i, j)$ is an element of the population matrix. Caswell (1985) describes the derivation of sensitivity in detail and the reader is referred to his work for more detailed descriptions of the analytic procedures. Briefly summarized, the sensitivity of λ to changes in $P(i, j)$ is proportional to the product of $v(i)$ and $w(j)$, where v and w are the left and right eigenvectors of the population matrix.

Elasticity

Weinberg et al. (1986) have defined elasticity which adjusts sensitivity for the initial values of λ and $P(i, j)$. It depicts the percent change in λ as a function of the percent change in $P(i, j)$. De Kroon et al. (1986) report that the elasticities of the population matrix sum to 1.0. Thus elasticity may be interpreted as the proportion of population growth described by a given element of the population matrix.

Non-equilibrium sensitivity analysis

Sensitivity and elasticity under non-equilibrium conditions were modeled in two ways. Firstly sensitivity was determined under conditions analogous to environmental changes that regularly alter the transition probabilities in the population matrix. For a given size class structure and a given population matrix, growth rates were calculated by iteratively multiplying the size class structure by the population matrix. Sensitivity was calculated by comparing the growth rate of that population to growth of a parallel population in which one of the transition probabilities was changed by 0.01. Sensitivity was calculated as $\Delta\lambda/\Delta P(i, j)$. Elasticities were calculated by adjusting the sensitivity to the transition probability and growth rate. Because the population structure changes continuously on its approach to the stable size structure, sensitivities and elasticities were calculated for each element of the population matrix for each year over 50 years.

A non-equilibrium sensitivity analysis was also conducted under conditions analogous to sudden shifts in population structure, such as those which occur with sudden inputs of larvae or fragments. In these calculations the growth of a single cohort was modelled using an initial population structure of 1000 small recruits. Growth rate was determined by comparing initial and final population size after iteratively multiplying the population structure by the population matrix. Calculated growth rates were cumulative growth rates averaged over periods of 1–50 years. Sensitivity was determined by comparing the cumulative population growth rates to those observed using an altered population matrix. As in the other non-equilibrium case the population was examined over 50 iterations (i.e. years). Growth rates, sensitivity and elasticity were calculated at each iteration, but were calculated comparing the population to the initial population, not to the population of the preceding year. The calculated values, therefore, characterize the cumulative effect of the non-equilibrium event averaged over the life of the cohort.

Table 1. Transition matrices characterizing *Plexaura* A colony growth at Korbiski Reef

Year	Population matrix		
1984	0.434	0.083	0.050
	0.023	0.500	0.075
	0.023	0.417	0.875
1985	0.447	0.163	0.000
	0.029	0.488	0.114
	0.000	0.163	0.867
1986	0.545	0.066	0.023
	0.091	0.665	0.000
	0.000	0.199	0.906
1987	0.243	0.045	0.023
	0.122	0.454	0.023
	0.000	0.045	0.845
Average year	0.468	0.116	0.029
	0.070	0.541	0.052
	0.006	0.186	0.881

Results

The *Plexaura* A data and the resultant population matrices are presented in Table 1. Population matrices were determined for each of four years and for an "average" year. Column sums of elements for the 0–9.9 and 10–19.9 cm colonies are equal to annual survivorship, and it is apparent that small colonies had lower survivorship than the other two size classes. The smallest colonies also had much lower probabilities of growing into larger colonies than intermediate size colonies. Consequently small colonies can spend many years in the smallest size class, which when coupled with the low survivorship reduces the probability of a small colony ever reaching the largest size class. The equilibrium growth rates and size class structures are presented in Table 2. Those values are dramatically different from the observed values. The difference between predicted and observed growth rates can be divided into two components, effects due to the population size class structure, and the recruitment of colonies not predicted by the closed model. The effects of size class structure are seen in the single-year growth rate predictions (Table 2). Those growth rates are lower than the equilibrium prediction due to the large number of colonies in the non-reproductive size classes. At equilibrium a greater percentage of the colonies would be reproductive and consequently the growth rate would be greater. The second effect, the higher than predicted numbers of small recruits, directly increased growth, but also maintained the skewed size structure throughout the 4 years of the study. Those "additional" recruits presumably came from outside the transect (i.e. the population on the transect was not closed). Despite these errors the equilibrium predictions correctly differentiated years with high growth from years with low growth. However, the equilibrium model did not accurately predict absolute growth rates or the size class structure.

The equilibrium sensitivity analysis (Table 3) indicates that transition probabilities leading to the development or maintenance of colonies > 20 cm have the greatest

Table 2. Size structure and growth rate of the *Plexaura* A population at Korbiski Reef.

Population structure						
Size Class					Popula- tion Size	Growth rate
0–9.9 cm			10–19.9 cm	> 20 cm		
Observed values						
June	1984	0.38	0.15	0.46	91	1.13
April	1985	0.41	0.14	0.46	101	1.08
June	1986	0.45	0.14	0.41	111	1.08
May	1987	0.44	0.18	0.38	119	0.67
May	1988	0.29	0.23	0.49	80	
Average		0.39	0.17	0.44		0.99
Equilibrium predictions						
June	1984	0.09	0.26	0.65		1.05
April	1985	0.08	0.22	0.71		0.92
June	1986	0.08	0.23	0.70		0.98
May	1987	0.04	0.07	0.89		0.89
Average		0.11	0.31	0.58		0.99
Single year predictions						
June	1984					0.87
April	1985	0.23	0.22	0.55		0.76
June	1986	0.27	0.18	0.55		0.46
May	1987	0.32	0.21	0.48		0.89
June	1988	0.20	0.24	0.56		

Table 3. Sensitivity and elasticity analysis of the *Plexaura* A population model. Each value in the matrices is the sensitivity or elasticity of the corresponding value of the population matrix

Year	Sensitivities			Elasticities		
1984	0.001	0.020	0.050	0.003	0.002	0.003
	0.080	0.236	0.579	0.002	0.236	0.579
	0.105	0.308	0.757	0.002	0.122	0.633
1985	0.002	0.007	0.021	0.001	0.001	0.000
	0.037	0.106	0.345	0.001	0.056	0.048
	0.096	0.275	0.892	0.000	0.048	0.843
1986	0.012	0.036	0.110	0.007	0.002	0.003
	0.059	0.173	0.532	0.006	0.117	0.051
	0.091	0.265	0.815	0.000	0.054	0.761
1987	0.001	0.002	0.020	0.000	0.000	0.001
	0.005	0.008	0.106	0.001	0.004	0.003
	0.043	0.076	0.991	0.000	0.004	0.987
Average year						
	0.011	0.030	0.056	0.005	0.004	0.002
	0.066	0.188	0.347	0.005	0.103	0.080
	0.153	0.433	0.802	0.001	0.081	0.720

effect on population growth. The importance of the large colonies can be attributed to the fact that all forms of reproduction are restricted to this class and to the predominance of large colonies in the stable size class structure. The importance of survival of the largest colonies is found consistently through all 4 years of the study. During 1987, a year of increased mortality, the sensitivity of survivorship of the > 20 cm colonies increased while all of the other sensitivities decreased. However, even in

Table 4. Elasticities calculated under non-equilibrium conditions following changes in the transition probability. Each value in the matrix is the elasticity for the corresponding value of the population matrix

1984	0.192	0.015	0.028
	0.010	0.088	0.118
	0.010	0.074	0.466
1985	0.239	0.030	0.000
	0.016	0.089	0.078
	0.000	0.030	0.520
1986	0.291	0.011	0.013
	0.049	0.114	0.045
	0.000	0.034	0.443
1987	0.173	0.013	0.015
	0.087	0.137	0.017
	0.000	0.014	0.544
Average Year	0.253	0.020	0.018
	0.038	0.094	0.111
	0.003	0.032	0.432

1987 the relative ranking of the sensitivity values remained constant. The elasticities again indicate that changes affecting growth into or survival of the largest size class is most important. Under average conditions 72% of population growth is controlled by maintenance of the largest colonies.

In the non-equilibrium analysis (Table 4) elasticity of survivorship in the smallest size class changes from one of the lowest values to the second highest. This same effect is observed using the transition probabilities and initial size structure of each of the four years of observation. Contrary to the prediction of the equilibrium solu-

tion there is the potential for strong selection on small-colony survivorship.

The equilibrium sensitivity values incorporate both the direct effects of changing the transition probability and the indirect effect of changing the stable size structure of the population. This latter effect is not incorporated in the non-equilibrium analysis. In the non-equilibrium analysis the sensitivity of the population to change in any given element of the population matrix was equal to the relative size of the affected size class (i.e. the sensitivities 1 year after the change in the population matrix were exactly equal to the observed size class abundances presented in Table 2). This effect is obscured somewhat when elasticity is calculated (Table 4). Elasticity shows that the effects of the perturbation depends on both the magnitude of the affected transition probability and the size class distribution. Table 5 shows how those elasticities change after 5, 10 and 15 years. As noted, the first year values are controlled by the initial size class distribution. In the subsequent years the elasticities are influenced by the cumulative effects of changing size class. The negative values observed in some of the elasticities in Table 5 are generated when the population with the altered population matrix approaches its stable size class structure more slowly than the unaltered population. The altered population has more non-reproductive colonies than the unaltered population and at that point in time has lower growth rates. After 10–15 years the stable age structure is reached in both the unaltered and altered populations and the sensitivities and elasticities approach the equilibrium values presented in Table 3.

The non-equilibrium effects are averaged over multiple years in the analysis in Table 6. In this analysis the sensitivity and elasticity of cumulative growth rate is presented instead of the single-year change in growth

Table 5. Elasticities calculated under non-equilibrium conditions at different times following changes in the transition probability

Time	Transition probability								
	(1,1)	(1,2)	(1,3)	(2,1)	(2,2)	(2,3)	(3,1)	(3,2)	(3,3)
1984									
1	0.192	0.015	0.028	0.010	0.088	0.118	0.010	0.074	0.466
5	−0.018	0.003	0.004	0.002	0.117	0.125	0.003	0.123	0.644
10	0.002	0.002	0.003	0.002	0.113	0.121	0.002	0.122	0.635
15	0.003	0.002	0.003	0.002	0.113	0.121	0.002	0.122	0.635
1985									
1	0.239	0.030	0.000	0.016	0.089	0.078	0.000	0.030	0.520
5	−0.038	0.005	0.000	0.000	0.076	0.062	0.000	0.051	0.847
10	−0.004	0.001	0.000	0.001	0.058	0.049	0.000	0.049	0.847
15	0.001	0.001	0.000	0.001	0.058	0.048	0.000	0.049	0.845
1986									
1	0.291	0.011	0.013	0.049	0.114	0.045	0.000	0.034	0.443
5	−0.029	0.005	0.006	0.010	0.159	0.055	0.000	0.063	0.738
10	−0.003	0.002	0.004	0.005	0.124	0.051	0.000	0.056	0.767
15	0.006	0.002	0.003	0.005	0.119	0.050	0.000	0.054	0.766
1987									
1	0.173	0.013	0.015	0.087	0.137	0.017	0.000	0.014	0.544
5	−0.021	−0.003	0.003	−0.012	−0.026	0.007	0.000	0.009	1.044
10	−0.001	−0.000	0.001	−0.000	−0.003	0.004	0.000	0.004	0.996
15	0.000	0.000	0.001	0.001	0.004	0.003	0.000	0.004	0.988

Table 6. Sensitivities and elasticities of the transition matrix calculated on the basis of cumulative growth over 50 years. Each value in the matrices is the sensitivity or elasticity of the corresponding value of the population matrix

Sensitivity			Elasticity		
0.051	0.042	0.069	0.025	0.005	0.003
0.277	0.200	0.312	0.021	0.114	0.075
0.597	0.431	0.691	0.004	0.085	0.650

rate. When averaged over many years the sensitivities and elasticities are very similar to the equilibrium conditions. As in all of the analyses, survival of the largest colonies has the greatest effect on population growth. Transitions affecting movement into or out of the largest size class are next in importance. Survivorship of small and intermediate size colonies have somewhat greater elasticities than predicted in the equilibrium solution. Production of small colonies, $P(1,3)$, had the smallest effect on population growth, and survivorship of the large colonies, $P(3,3)$ had the largest effect on population growth.

Discussion

Sensitivity and elasticity analyses based on equilibrium solutions assume steady state population structure and unchanging environmental conditions. Although the San Blas Islands are not subject to catastrophic events such as hurricanes, the *Plexaura* A transition probabilities changed over the course of the study. Furthermore, the population structure at Korbiski changed each year and always differed from the predicted stable size class distribution. Temporal change in survival and growth rates, the recruitment of fragments from outside of the transect and the occasional recruitment of larvae all violate the assumptions of the model. The *Plexaura* A population at Korbiski, like many benthic populations, was not at equilibrium.

Two types of non-equilibrium condition have been analyzed in this study. In one case sensitivities were calculated for the populations using the population matrices and size structures observed at Korbiski. This is analogous to the case of continuously changing life history parameters or continuous inputs of recruits which keep the population from ever approaching its stable size structure. Under these circumstances sensitivity and elasticity are strongly influenced by the size structure of the population. Among *Plexaura* A at Korbiski the size structure was skewed from the stable size structure by having more small colonies than predicted. That had the effect of increasing the elasticity of the survival of the smallest colonies [$P(1,1)$ in Table 4]. The comparison of equilibrium and non-equilibrium sensitivity analyses of the *Plexaura* A population model show that the equilibrium solution accurately characterizes the sensitivity of most of the elements of the population matrix but tends to underestimate the importance of survival of the smallest colonies. In this case there can be greater selection

on small colonies than predicted from the equilibrium sensitivity analysis.

The second type of non-equilibrium result is that which occurs when a sudden influx of individuals occurs (Table 6) or when there is a sudden change in transition probabilities which then return to their pre-disturbance values. In both of these cases the population rapidly approaches its stable size class distribution and the sensitivity and elasticities approach those values predicted by the equilibrium analysis. Thus among populations subject to rare pulses of recruitment, selection pressures do not differ very much from the equilibrium prediction. Episodic recruitment events, which are frequently observed among benthic invertebrates, can only change the opportunity for selection if they occur often. In *Plexaura* A such recruitment would have to occur every 5–10 years to notably affect sensitivities or elasticities. Catastrophic events such as hurricanes or bleaching could have a similar temporary effect on the population size structure. However, changes in competitive interactions or grazing (Knowlton et al. 1990) often co-occur with changes in population density. Thus mass mortalities may have greater and longer lived effects than predicted by the initial change in population size structure alone.

Although larval reproduction was not explicitly included in the model it is also possible to compare the potential effects of larval and vegetative reproduction on *Plexaura* A population growth. Larval reproduction, if included in the model, would contribute to $P(1,3)$. $P(1,3)$ had sensitivity and elasticity values which were lower than almost all of the other transition probabilities. $P(2,3)$, the transition probability which contains most vegetative propagation, had an elasticity an order of magnitude greater than $P(1,3)$. Thus any decrease in larval production if accompanied by an equivalent increase in vegetative propagation would lead to an increase in population growth. The sensitivity and elasticity analyses suggest that the continued maintenance of larval reproduction may require consideration of factors not included in the model, such as dispersal. [Since *Plexaura* A larvae are possibly parthenogenetic in origin (Brazeau and Lasker 1989), comparison of larval and vegetative reproduction may be independent of the genetic costs and benefits of sex.] Although the relative costs of vegetative propagation and larval reproduction have not yet been incorporated in the *Plexaura* A model, the analysis suggests the magnitude of the differences required to maintain larval reproduction. Caswell and Werner (1978) reached a somewhat analogous conclusion when they conducted a transient analysis of growth in the plant *Dipsacus sylvestius*. They suggested that under non-equilibrium settings selection might operate on traits such as seed production (i.e. dispersal) or rosette size (i.e. competitive ability) instead of local population growth.

In their transient analysis of a *Dipsacus* life history analysis Caswell and Werner (1978) concluded that short term and long term predictions closely corresponded with one another. In this study the short-term, non-equilibrium, predictions were often similar to the long-term, equilibrium, solution. However, as Caswell and

Werner (1978) cautioned, such similarities are probably specific to the models that have been examined. Investigators interested in exploring evolutionary predictions using population models must also consider non-equilibrium conditions in their analyses.

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